

Resonant Reactions and Neutron-Induced Processes

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Two Classes of Nuclear Reactions

Non-resonant reactions (one-step process):

- Direct capture, stripping, pickup, charge exchange, Coulomb excitation/dissociation
- Cross section $\sigma(E)$ is a **smooth** function of projectile energy E
- Occurs at essentially any projectile energy; governed by an electromagnetic operator matrix element

Resonant reactions (two-step process):

- Step 1: projectile x + target A *form* a discrete excited state of the compound nucleus
- Step 2: that excited state *decays* to the final channel
- Happens only when $E_{\text{proj}} + Q$ matches a discrete excited-state energy of the compound nucleus $\Rightarrow \sigma(E)$ varies **dramatically** at specific resonance energies E_R

Generic reaction: $x + A \rightarrow (\text{compound nucleus})^* \rightarrow B + \gamma$ (or b)

Damped-Oscillator Analogy

A resonant reaction behaves like a driven, damped oscillator:

- Response (cross section) is maximal when driving frequency matches the system's *Eigen frequency* $\leftrightarrow E = E_R$
- Oscillator strength $\sim$ product of the two partial widths, $\Gamma_a \Gamma_b$
- Damping factor $\leftrightarrow$ total width Γ of the nuclear reaction
- Lifetime of the compound state τ_0 related to Γ by Heisenberg's uncertainty principle:

$$\Gamma \tau_0 = \hbar$$

Partial widths:

- Γ_a – *formation* width (probability of forming the compound state via channel *a*)
- Γ_b – *decay* width (probability of decay via channel *b*: particle or γ emission)
- Total width: $\Gamma = \Gamma_a + \Gamma_b$

Maximum Geometrical Cross Section

Maximum cross section a projectile can present to a target, per partial wave l :

- Reduced de Broglie wavelength: $\lambda = \hbar/p$
- Orbital angular momentum: $L = bp = l\hbar \Rightarrow b = l\lambda$ (impact parameter)
- Maximum cross section for partial wave l :

$$\sigma_{l,\max} = \pi (b_{l+1}^2 - b_l^2) = (2l + 1) \pi \lambda^2$$

Including spin (statistical factor):

$$\omega = \frac{2J + 1}{(2J_1 + 1)(2J_2 + 1)}$$

where J = compound-nucleus spin, J_1, J_2 = spins of projectile and target. Reduces to $(2l + 1)$ for spinless particles. **Identical particles:** cross section doubled via the Kronecker factor $(1 + \delta_{12})$.

The Breit–Wigner Formula

Combining the geometrical cross section, statistical factor, identical-particle correction, and the resonance (Lorentzian) energy dependence:

$$\sigma(E) = \pi\lambda^2 \frac{2J+1}{(2J_1+1)(2J_2+1)} (1 + \delta_{12}) \frac{\Gamma_a \Gamma_b}{(E - E_R)^2 + (\Gamma/2)^2}$$

- E_R : resonance energy; $\Gamma = \Gamma_a + \Gamma_b$: total width
- Valid for **isolated** resonances: $\Gamma \ll$ separation between compound-nucleus excited states
- Valid for **narrow** resonances: $\Gamma \ll E_R$, so Γ is effectively constant across the resonance width
- $\pi\lambda^2$ plays the role of the geometric cross section πr^2 “seen” by the projectile

Reaction Rate: Setting Up the Integral

General stellar reaction rate (Maxwell–Boltzmann averaged):

$$\langle \sigma v \rangle = \sqrt{\frac{8}{\pi \mu}} \frac{1}{(kT)^{3/2}} \int_0^\infty E \sigma_{BW}(E) e^{-E/kT} dE$$

For a **narrow** resonance, $e^{-E/kT}$ varies negligibly across the resonance width, so it can be pulled out at $E = E_R$:

$$\langle \sigma v \rangle \approx \sqrt{\frac{8}{\pi \mu}} \frac{1}{(kT)^{3/2}} E_R e^{-E_R/kT} \int_0^\infty \sigma_{BW}(E) dE$$

The remaining task: evaluate $\int_0^\infty \sigma_{BW}(E) dE$ analytically using the Breit–Wigner form.

Resonance Strength $\omega\gamma$

The Breit–Wigner integral evaluates to:

$$\int_0^\infty \sigma_{BW}(E) dE = 2\pi^2 \lambda^2 \omega \frac{\Gamma_a \Gamma_b}{\Gamma}$$

Define the **resonance strength**:

$$\omega\gamma \equiv \omega \frac{\Gamma_a \Gamma_b}{\Gamma}, \quad \gamma \equiv \frac{\Gamma_a \Gamma_b}{\Gamma}$$

At exact resonance, the peak cross section is

$$\sigma(E_R) = 4\pi \lambda_R^2 \omega \frac{\Gamma_a \Gamma_b}{\Gamma^2} = \sigma_R$$

and the area under the resonance curve is (total width) $\times$ (height):

$$\int_0^\infty \sigma_{BW}(E) dE = \frac{\pi}{2} \Gamma \sigma_R$$

Why it matters: $\omega\gamma$ is an **experimentally measurable** quantity that fully characterizes a narrow resonance's contribution to the reaction rate.

Reaction Rate for a Narrow Resonance

Substituting back, the stellar reaction rate per particle pair for a single narrow, isolated resonance:

$$\langle \sigma v \rangle = \left(\frac{2\pi}{\mu kT} \right)^{3/2} \hbar^2 (\omega\gamma)_R e^{-E_R/kT} f$$

- $(\omega\gamma)_R$: resonance strength at E_R (measured experimentally)
- $e^{-E_R/kT}$: Maxwell–Boltzmann factor – the **Gamow peak is replaced by the resonance peak** at E_R
- f : electron screening factor (included for completeness; negligible unless stellar density is very high)

Striking feature: no cross-section/integral term remains. To get the reaction rate, only *three* measured quantities are needed: resonance strength, width, and resonance energy. **Many narrow resonances:**

$$\langle \sigma v \rangle = \left(\frac{2\pi}{\mu kT} \right)^{3/2} \hbar^2 \sum_i (\omega\gamma)_i e^{-E_{R,i}/kT} f$$

Formation vs. Decay Width Near/Below the Coulomb Barrier

Consider radiative capture $x(A, \gamma)B$, with formation width Γ_x and decay (gamma) width $\Gamma_\gamma \lesssim 1$ eV (always very small). **Case 1 – E_R near the Coulomb barrier V_c :**

$$\Gamma_x \sim \text{MeV} \gg \Gamma_\gamma \Rightarrow \omega\gamma = \frac{\Gamma_x \Gamma_\gamma}{\Gamma} \approx \Gamma_\gamma$$

Case 2 – E_R far below V_c :

$$\Gamma_x \ll \Gamma_\gamma \Rightarrow \omega\gamma = \frac{\Gamma_x \Gamma_\gamma}{\Gamma_\gamma} \approx \Gamma_x$$

Since Γ_x contains the barrier-penetration term, it falls off **extremely rapidly** as E_R decreases below $V_c \Rightarrow$ resonance strength drops very fast with decreasing energy. From the reaction rate $\langle \sigma v \rangle \propto \sum_i (\omega\gamma)_i e^{-E_R/kT}$: resonances with E_R near kT dominate, since this is the only place E_R and kT appear comparably.

Worked Example: $^{14}\text{N}(\text{p},\gamma)^{15}\text{O}$

Take $T_6 = 30$ (i.e. $T = 3 \times 10^7$ K, so $kT \approx 2.6$ keV).

- Compare a **low-energy** resonance at $E_R = 10$ keV with a **high-energy** resonance at $E_R = 2$ MeV
- Ratio of reaction rates (low-energy : high-energy) $\approx 10^{312}$

Conclusion: a small difference in resonance energy changes the mean lifetime of nuclei – and hence the star's properties – **dramatically**. Knowing the precise location and strength of *low-energy* resonances is far more important for stellar burning than knowing high-energy ones. (*Reference: Rhodes & Rodney, for the full numerical calculation.*) **Special case – neutron-induced resonant capture:** no Coulomb barrier in the formation channel, so $\Gamma_a \gg \Gamma_b$ generally, and $\omega\gamma \approx \omega\Gamma_b$ holds for *nearly all* resonance energies (not just low ones).

Number of Nuclei in an Excited Compound State

Question: if the compound nucleus is formed in a long-lived excited state (e.g. a metastable/isomeric level), how many such nuclei are available to react further *before* decaying to the ground state? **Production rate**

(proportional to number of nuclei in resonance state; at high E_R , $\Gamma_a \gg \Gamma_b \Rightarrow \Gamma \approx \Gamma_a \Rightarrow \omega\gamma \approx \omega\Gamma_b$):

$$r_{12} = \frac{N_1 N_2}{1 + \delta_{12}} \langle \sigma v \rangle = \frac{N_1 N_2}{1 + \delta_{12}} \left(\frac{2\pi}{\mu k T} \right)^{3/2} \hbar^2 \omega \Gamma_b e^{-E_R/kT}$$

Decay rate (Γ_b sets the lifetime τ_b of the N_{12} excited nuclei, via $\Gamma_b \tau_b = \hbar$):

$$r_{12} = \frac{N_{12}}{\tau_b} = \frac{\Gamma_b N_{12}}{\hbar}$$

The Saha Equation

Equating production rate = decay rate (equilibrium) and solving for N_{12} :

$$N_{12} = \frac{N_1 N_2}{1 + \delta_{12}} \left(\frac{2\pi}{\mu k T} \right)^{3/2} \hbar^3 \omega e^{-E_R/kT}$$

This is the (Meghnad) **Saha equation** – valid for *any* reaction once production/decay equilibrium is established. It gives the number of nuclei participating in the collision while in an excited compound state. **Classic example**

– **triple- α reaction:**

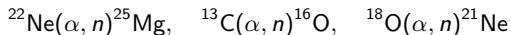


$$\frac{N({}^8\text{Be})}{N({}^4\text{He})} \sim 10^{-10}$$

Tiny, but **sufficient** to bridge the mass-5 and mass-8 gaps and allow ${}^8\text{Be} + \alpha \rightarrow {}^{12}\text{C}$ – the route by which carbon is synthesized.

Why Neutrons Are Different

Neutrons carry no charge $\Rightarrow$ no Coulomb barrier to penetrate $\Rightarrow$ **no Gamow peak**. **Origin of free neutrons in stars** (mainly during helium burning):



These neutrons synthesize nearly all elements **beyond the iron peak** (proton/alpha-induced reactions dominate synthesis below iron). **Lifecycle of a free neutron:**

- Born with MeV-scale energy
- Thermalized via elastic scattering in $\sim 10^{-11}$ s; velocities then follow a Maxwell-Boltzmann distribution
- Free-neutron lifetime ≈ 10 minutes (vs. $\sim 10^9$ years for free protons) – short, but enormously longer than thermalization time
- Laboratory analogue: moderators (water, paraffin/wax, light hydrogen-rich materials) thermalize fast neutrons, exactly as in a nuclear reactor

Cross Section: General Form and the $1/v$ Law

Generic neutron-induced reaction $X(n, a)Y$, treated as a two-step process:

$$\sigma_n = \lambda^2 \Gamma_n(E_n) \Gamma_a(Q + E_n) \approx \lambda^2 \Gamma_n(E_n) \Gamma_a(Q)$$

since $E_n \ll Q$ for thermalized neutrons ($E_n \sim 30$ keV vs. $Q \sim$ MeV) $\Rightarrow$ exit-channel width depends essentially only on the (constant) Q -value. ***s*-wave neutrons ($l = 0$):** $\lambda^2 \propto 1/v_n^2$ and $\Gamma_n \propto v_n$ (entrance-channel width $\propto$ neutron velocity), so

$$\sigma_n \propto \frac{1}{v_n^2} \cdot v_n = \frac{1}{v_n} \quad \Rightarrow \quad \boxed{\sigma_n \propto \frac{1}{v_n}} \quad (\text{the “1/v law”})$$

This smooth $1/v$ dependence is why these are called **non-resonant**. It applies not only to (n, γ) but also to $(n, \text{charged particle})$ exit channels, since charged-particle transmission coefficients are constant at their (MeV-scale) exit energies. **Consequence:** $\langle \sigma v \rangle$ is **constant** – independent of temperature – and can in principle be determined from σ measured at *any* single velocity.

Breakdown of the $1/v$ Law

The $1/v$ law fails when:

- 1 s -wave neutrons are *not* low-energy, or
- 2 Higher partial waves ($l \geq 1$) contribute to σ

For higher partial waves, $\Gamma_l \sim E^{l+1/2}$, so with $\lambda^2 \propto 1/v_n^2 \propto 1/E$:

$$\sigma_l \sim \frac{1}{v^2} \Gamma_l \sim E^{l-1/2}$$

l	Partial width Γ_l	Cross section σ_l
0	$E^{1/2}$	$E^{-1/2}$
1	$E^{3/2}$	$E^{1/2}$
2	$E^{5/2}$	$E^{3/2}$

The reaction rate integrand $E \sigma(E) e^{-E/kT} \sim E^{l+1/2} e^{-E/kT}$ defines an **effective energy window**: its peak shifts to higher energy as l increases (curves normalized at $kT = 30$ keV).

Energy-Dependent Cross Section: Taylor Expansion

When $\langle\sigma v\rangle$ is *not* constant (i.e. the pure $1/v$ law fails), expand $\sigma v = S(\sqrt{E})$ in a Taylor series about $E = 0$:

$$\sigma v = S(\sqrt{E}) \approx S(0) + \dot{S}(0)\sqrt{E} + \frac{1}{2}\ddot{S}(0)E$$

so the energy-dependent cross section becomes

$$\sigma(E) \approx \sqrt{\frac{\mu}{2E}} \left(S(0) + \dot{S}(0)\sqrt{E} + \frac{1}{2}\ddot{S}(0)E \right)$$

This mirrors the astrophysical S -factor expansion used for charged-particle non-resonant reactions, adapted here for neutron-induced processes.

Burning Stages as a Reaction Network

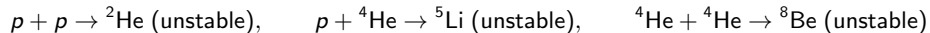
Real stellar evolution requires treating nuclear reactions as an **interconnected network**, not isolated processes.

Guiding principle: nuclides with the *lowest* Coulomb barrier burn first (tunneling probability is highest) $\Rightarrow$ H and He ignite before heavier fuels.

- After the lowest- V_c fuel is depleted in the core, its burning does **not** vanish – it continues in a surrounding **shell**
- Total stellar mass determines how many burning stages occur: $\text{H} \rightarrow \text{He} \rightarrow \text{C} \rightarrow \text{Ne} \rightarrow \text{O} \rightarrow \text{Si} \rightarrow \dots$
- Beyond charged-particle/neutron capture: *s*-process, *r*-process, (*p*-process) responsible for heavy-element nucleosynthesis

Hydrogen Burning: The First Stage

Most probable initial reactions among protons/helium (all immediate products are unstable):



None of these is a viable end-point on its own – their instability is precisely *why* stars need alternate reaction pathways (e.g. the *pp*-chain via β^+ decay, or bridging via the Saha-equation population of ${}^8\text{Be}$ in the triple- α process) to stabilize and proceed. **Stellar structure picture:** as the core exhausts its lowest- V_c fuel and gravitational contraction raises the central temperature, central density and burning-phase duration evolve through $\text{H} \rightarrow \text{He} \rightarrow \text{C} \rightarrow \text{Ne} \rightarrow \text{O} \rightarrow \text{Si}$ stages, each providing thermal pressure that (temporarily) halts further contraction.

Summary I: Resonant Reaction Formalism

- Resonant reactions: two-step process (formation Γ_a , decay Γ_b) $\rightarrow$ Breit–Wigner cross section

$$\sigma(E) = \pi \lambda^2 \omega (1 + \delta_{12}) \frac{\Gamma_a \Gamma_b}{(E - E_R)^2 + (\Gamma/2)^2}$$

- For narrow, isolated resonances, the stellar reaction rate collapses to a clean closed form set by three measurables: E_R , Γ , and the resonance strength $\omega\gamma$
- **Low-energy** resonances near kT overwhelmingly dominate stellar reaction rates (illustrated: $\sim 10^{312}$ ratio for $^{14}\text{N}(p,\gamma)^{15}\text{O}$)
- Saha equation gives the population of nuclei in long-lived excited compound states – essential for bridging the ^8Be mass gap in the triple- α process

Summary II: Neutron-Induced Reactions and Burning

- Neutron-induced reactions: no Coulomb barrier $\Rightarrow$ no Gamow peak; s -wave capture follows the smooth $1/v$ law, $\sigma_n \propto 1/v_n$
- $1/v$ law breaks down for non-low-energy s -wave neutrons or when higher partial waves ($l \geq 1$) contribute, shifting the effective energy window
- Free neutrons (lifetime ~ 10 min) produced via (α, n) reactions during He-burning drive nucleosynthesis of elements **beyond the iron peak**
- Stellar burning proceeds as a network: lowest-Coulomb-barrier fuel burns first (H, then He, ...); exhausted fuel keeps burning in a shell; total mass sets how many stages (H/He/C/Ne/O/Si) a star undergoes